New experiments on 25 controlled models (DataDecide)

1 Overview

The paper claims that **how often numbers of each size appear in pretraining data** (α) is related to **how compressed the model's internal number line is** (β). The reviewers mainly asked whether other differences between the models explain this, whether 6 corpora are enough, whether the result is robust, whether β is measured reliably, and how numbers were counted. All new experiments below use a set of models built to remove most of these doubts.

Terms used throughout.

- **DataDecide.** 25 language models released by Ai2. All have 1B parameters, the same architecture (16 layers), the same tokenizer, and were trained for the same number of steps on 100B tokens. *Only the training data (the "recipe") differs.* Recipes come in **families**: variants of one base corpus (filtered subsets, versions with a part removed, or mixes). The families are Dolma, DCLM, Falcon, FineWeb, C4, and DCLM+Dolma mixes.
- α **(data exponent).** Count how often each whole number n appears in the training text, $c(n)$. Larger numbers are rarer. α is the slope of $\log c(n)$ against $\log n$. It is negative; a *less negative* α means large numbers are relatively more common.
- β **(number-line compression).** We give the model prompts that end in a number, take its internal state (hidden state) at one layer, and find the main direction along which those states vary (the first principal component, PC1). Numbers line up along PC1 by size. β says how the space between magnitude groups changes from one power of ten to the next: $\beta = 1$ gives each power of ten equal space (a logarithmic line); $\beta < 1$ squeezes large numbers even more. *Higher* β *= less compression.*
- **Prediction.** Flatter α (large numbers more common) $\Rightarrow$ higher β. So we expect a *positive* correlation between α and β.
- **Layer choice.** The number line is read from one layer per model, chosen by a fixed rule from the paper: the layer where PC1 explains the most variance *and* orders the numbers best (score $\sqrt{\text{explained variance} \times |\text{order}|}$).
- **Order.** Spearman correlation between n and its position on PC1 (1 = numbers perfectly in order).
- **Pearson** r **/ Spearman** ρ**.** r measures a straight-line relation; ρ only asks whether the *ranking* agrees and is less sensitive to single extreme models. p is the probability of seeing a correlation this strong if there were no real relation.

Not yet done: E08 (tracking β over training checkpoints) and E09 (testing on new models, Paloma) are planned but not run. Nothing here shows *causation*; all results are correlations across models.

Table 1: The new experiments at a glance. Reviewer IDs refer to the AAAI reviews.

Exp.	Question	Main answer	Helps with
E01	Does the paper's result hold on 25 models that differ only in data?	Yes, moderately: $r = 0.56$ ($p = 0.003$), $\rho = 0.46$ ($p = 0.02$).	Confounding by architecture, tokenizer, size, training length; "only 6 corpora"; partial correlation for scale (H7EL, VRQv, a1nj).
E04	Is α counted on what the model actually saw?	Yes: exact counts of each model's own 100B training tokens; sampling noise is tiny.	How counts were made: sampling weights, repeated data, match to the checkpoint (VRQv Q3).
E05	Can other properties of the data explain β?	Partly: code/math content (only in Dolma) is a strong confound for the original α.	Partial correlations and confounds (a1nj). Honest limitation.
E06	Is β reliable when fitted from only 3 gaps?	Yes: 10 groups, 3 seeds; β is stable and agrees with the 3-gap β ($r = 0.96$). Layer choice is the fragile part.	β from few points, fit quality, stability (a1nj pt. 3); sensitivity to the layer.
E10	Should any model be excluded, by rules set blind to the result?	Only 4 models flagged; excluding them changes little. No valid exclusion strengthens the result.	Inclusion/exclusion criteria (H7EL, VRQv Q1).
E11	What if α is measured on the same numbers β is measured on?	$r = 0.82$, $\rho = 0.77$; holds without Dolma, within families, and when any model is removed.	Sensitivity, leave-one-out, Dolma influence, by-family analysis, uncertainty (VRQv Q2, H7EL, a1nj Q1).

2 E01: The paper's method on 25 controlled models

Purpose. In the paper, models differ in architecture, tokenizer, size *and* data, so any of these could explain β. DataDecide fixes everything except the data.

What we did. Exactly the paper's protocol.

- Prompts `a=a,b=b,c=c,n=`: three examples of repeating a number, then the model must repeat n. Only prompts the model answers correctly are kept.
- n comes from 4 magnitude groups: 1–39, 80–119, 980–1019, 9980–10019 (40 prompts each). The 4 group positions on PC1 give **3 gaps**, from which β is fitted.
- The layer is chosen on one set of prompts (seeds 42–44) and β is measured on fresh prompts (seeds 45–47), so the layer is not tuned on the data it is evaluated on.
- α is the paper's α (slope over all numbers 1–10000), counted exactly (E04).

Results. Per-model values are in Table 10.

- Every model has a clear number line: order = 0.90–0.96.
- β ranges from 0.62 to 1.36. Dolma-trained models are least compressed.
- Mixing DCLM with more Dolma raises β step by step: 0.67, 0.84, 0.95, 1.11 for 0, 25, 50, 75% Dolma.

What it addresses.

Table 2: α vs β: the paper versus the 25 DataDecide models.

Data	Points	Pearson r (p)	Spearman ρ (p)
Paper (corpus level, after inclusion audit)	6 corpora	0.81 (0.061)	0.60 (0.24)
DataDecide, E01	25 models	0.56 (0.003)	0.46 (0.022)

- **Confounding by model design:** architecture, tokenizer, size and training length are identical, so they cannot explain the relation.
- **Too few points:** 25 models instead of 6 corpora; the correlation is weaker than the paper's but now statistically significant.
- **"Partial correlation controlling for scale":** unnecessary here, since scale is the same for all models.
- **Limitation:** the correlation is moderate and largely driven by the Dolma family (see E05, E11).

3 E04: Exact number counts of each model's training data

Purpose. Make sure α describes the exact text each model was trained on, not an approximation.

What we did. For each model we rebuilt its training stream: the same files, the same shuffling, the same 100B tokens up to the released checkpoint (step 69,369). For the 3 recipes smaller than 100B tokens, training repeats part of the data; we counted it the same way, so repeats are counted as often as the model saw them. We then counted every whole number from 0 to 10,000. (Earlier checks, E02 and E03, confirmed that counting a random sample gives the same α as counting the full corpus.)

Table 3: How much α changes with the exact sample of data, compared with differences between recipes.

Comparison	Change in α
C4: 5 different 100B training streams vs the full 138B-token corpus	at most 0.0003
Dolma 1.7: 5 almost non-overlapping 100B streams (5.8% shared)	s.d. 0.001
Same recipe, two different data-order seeds	at most 0.0013
Between the 25 recipes	-1.43 to -1.76 (range 0.33)

Results. Differences between recipes are 75–900 times larger than the sampling noise, so α is a real, stable property of each recipe. Each model saw 1.0–2.0 billion numbers.

What it addresses. How counts were obtained (VRQv Q3): no sampling weights are needed (we count the actual training stream), repeated data is counted as many times as it was seen, and the counts match the evaluated checkpoint exactly.

4 E05: Other properties of the data (confounders)

Purpose. Recipes differ in more than α. We checked whether other properties explain β better.

What we did. For each recipe we measured *number density* (numbers per 1000 tokens) and the share of *code* and *math* documents. We correlated each with β, then computed *partial correlations*: the α–β correlation after removing what the other property explains. "Within families" means each family's average is subtracted first, so only differences between close variants of the same corpus remain.

Table 4: Correlation of each data property with β (25 models, original α).

Analysis	Pearson r with β (p)
α	0.56 (0.003)
Number density	0.47 (0.02)
Code share / math share	0.85 / 0.76
α, controlling for number density	0.50 (0.01)
α, controlling for number density and code share	−0.20 (0.36)
α, within families	0.28 (0.17)
α, without the 6 Dolma models	0.22 (0.36)

Results. Code and math documents exist only in the Dolma-based recipes, so "code share" is essentially "is it Dolma?". Once this is controlled, the original α no longer predicts β, and it is weak within families. In the DCLM→Dolma mixing series, β rises with the Dolma share ($\rho = 0.90$), but number density and code share rise with it too.

What it addresses. The confounding and partial-correlation requests (a1nj). It is an honest *limitation* for the original α, and the reason for E10 and E11.

5 E06: Is β reliable? (10 groups instead of 4)

Purpose. The paper fits β from only 3 gaps between 4 groups. Reviewers asked whether that is stable and how good the fit is.

What we did.

- 10 magnitude groups instead of 4, evenly spaced on a log scale (centres 10, 22, 46, 100, . . . , 10,000, each ±10%), giving 9 gaps.
- 100 correctly answered prompts per group, repeated with 3 random seeds (3,000 prompts per model).
- β is fitted with one smooth curve through all prompts, $\text{PC1} = a + s\,\beta^{\log_{10} n}$, instead of from group averages.
- We also report α vs β with *the same layer for every model*, to see how much the layer choice matters.

Results (reliability). β barely changes with new prompts, the curve fits well, and the paper's 3-gap β gives almost the same values. The fragile part is the *layer choice*: for 8 models the best and second-best layers score within 8% of each other, and switching between them can change β a lot (e.g. 0.81 at layer 9 vs 1.38 at layer 2 for one model). Because of these near-ties, α vs β is $r = 0.63$ ($p = 0.001$) at E01's layers but only 0.20 at E06's own choices.

Table 5: Reliability of β (25 models).

Check	Result
Change in β across the 3 prompt seeds (median s.d.)	0.015
Spread of β between models	0.55–1.38
Fit quality of the smooth curve (R^2, median and range)	0.89 (0.81–0.97)
Agreement with E01's 3-gap β, same layer (Pearson r)	0.96
Models whose chosen layer changes vs E01	8 of 25

Table 6: α vs β when every model is read at the *same* layer (E06 measurement, original α, $n = 25$ per layer).

		Median	Pearson		Spearman	
Layer	Order	β	r	p	ρ	p
1†	0.07	0.20	+0.35	0.090	+0.32	0.117
2	0.90	0.93	−0.05	0.799	−0.03	0.878
3	0.96	0.65	−0.35	0.084	−0.37	0.069
4	0.96	0.69	−0.00	0.987	−0.09	0.677
5	0.96	0.65	+0.51	0.010	+0.30	0.149
6	0.95	0.64	**+0.68**	0.0002	+0.56	0.004
7	0.95	0.64	**+0.67**	0.0003	+0.53	0.007
8	0.94	0.66	**+0.75**	0.00002	**+0.71**	0.00008
9	0.94	0.71	**+0.64**	0.0006	+0.54	0.005
10	0.94	0.72	**+0.61**	0.001	+0.45	0.023
11	0.94	0.70	**+0.66**	0.0004	**+0.59**	0.002
12	0.89	0.63	+0.48	0.016	+0.49	0.014
13†	0.44	0.53	−0.37	0.065	−0.14	0.504
14†	0.30	0.28	+0.26	0.214	+0.27	0.184
15†	0.30	0.21	−0.17	0.416	−0.20	0.349
16†	0.57	0.40	−0.35	0.085	+0.12	0.553

Order: median over models of how well PC1 orders the numbers. †Order below 0.85: no clear number line at that layer, so β there is not meaningful. **Bold**: significant after correcting for testing 16 layers ($p < 0.05/16 = 0.003$).

Results (fixed layer). In the middle layers (6–11), where most models' chosen layers lie, α and β are clearly related at every layer ($r = 0.61$–0.75). In early layers (2–4) there is no relation.

What it addresses. "β is estimated from few points; how stable is it, how good is the fit?" (a1nj point 3): stable, good fit, and the 3-gap β agrees with the 10-group β. Also shows how the result depends on the layer, and that comparing models at the same depth gives a clear relation.

6 E10: Blind audit of which models to exclude

Purpose. The paper excludes models using an inclusion audit. Reviewers asked how these choices affect the result. We asked: is there a *valid* reason to exclude any of the 25 models, decided without looking at the result?

What we did. 21 independent AI-agent auditors each checked one aspect of all 25 models (e.g. training health, how well the number line is measured, layer choice, tokenization, data repetition, leverage of individual models). None of them saw the α–β correlation. Each had to state its exclusion

rule and threshold first, and say whether a flag is a *measurement failure* (valid to exclude) or a *property of the data* (not valid to exclude, since that is what we study). Only then were the rules applied.

Table 7: The only models flagged as measurement problems.

Model	Reason
`dolma1_7-no-reddit`	Its β (1.36) does not replicate: 1.22 on other prompts, 1.13 in E06.
`dolma1_7`, `fineweb-edu`, `falcon-and-cc-qc-10p`	Two layers score almost equally but give β values more than 0.10 apart, so "β at the best layer" is ill-defined.

Results. With the original α: all 25 models give $r = 0.56$, $\rho = 0.46$; without the 4 flagged models $r = 0.50$, $\rho = 0.38$. No valid rule makes the result stronger. One auditor found that the original α mostly describes 4-digit numbers, while β only uses 4 small groups; this led to E11.

What it addresses. Inclusion/exclusion criteria (H7EL, VRQv Q1): exclusion rules were set before seeing the result, are explicit, and flag only 4 of 25 models. The conclusions do not depend on excluding models.

7 E11: α measured on the numbers that β uses

Purpose. β is measured only on 4 groups of numbers (1–39, 80–119, 980–1019, 9980–10019). The original α is a slope over *all* numbers 1–10000, so 90% of its points are 4-digit numbers (1000–9999) that β never sees. The two measures therefore describe different numbers. E11 measures α on the same numbers as β.

What we did.

- α_{old}: slope of $\log c(n)$ against $\log n$ over every number 1–10000 (as in the paper).
- α_{new}: for each of the 4 groups, take the median count of its numbers and the group's centre; α_{new} is the slope of log(median count) against log(centre), i.e. 4 points, one per group. The median keeps one unusually common number (e.g. 10000) from dominating a group. Counts stop at 10000, so the top group uses 9980–10000.
- Same 25 models, same β and layers as E01. No model is dropped.

Table 8: Example: why the two α can disagree. Median count of the numbers in each group.

Model	1–39	80–119	980–1019	9980–10000	α_{old}	α_{new}	β
`fineweb-edu`	13.4M	1.29M	51k	1.9k	−1.70	−1.37	**0.89**
`c4`	11.4M	0.93M	34k	1.5k	−1.56	−1.39	**0.62**

α_{old} says fineweb-edu's numbers become rare faster, so it should be *more* compressed; it is less compressed. On the probed numbers, fineweb-edu's counts actually drop more slowly, and α_{new} ranks the two correctly. The difference in α_{old} comes from 4-digit numbers between the probed groups.

Results.

- With α_{new} the relation is much stronger: $r = 0.82$ ($p < 10^{-6}$), $\rho = 0.77$, with no model removed.

Table 9: α vs β with both definitions of α. "Within families": each family's average removed first. $^*p < 0.05$.

		α_{old} (1–10000)		α_{new} (probed numbers)	
Models used	n	r	ρ	r	ρ
All 25	25	0.56*	0.46*	**0.82***	**0.77***
Without the 4 flagged models (E10)	21	0.50*	0.38	0.79*	0.70*
Without the 6 Dolma models	19	0.22	0.11	0.68*	0.66*
Without all 9 Dolma-containing models	16	−0.31	−0.33	0.43	0.53*
Without the 8 recipes with code or math	17	−0.38	−0.41	0.44	0.52*
Without the 7 DCLM models	18	0.59*	0.59*	0.86*	0.82*
Without the 6 Falcon models	19	0.63*	0.62*	0.83*	0.81*
Only the 8 base corpora (one per original corpus)	8	0.55	0.50	0.92*	0.90*
Within families	25	0.28	0.24	0.75*	0.75*
Remove one model at a time (range)	24	0.50–0.61	0.39–0.52	0.79–0.84	0.75–0.80

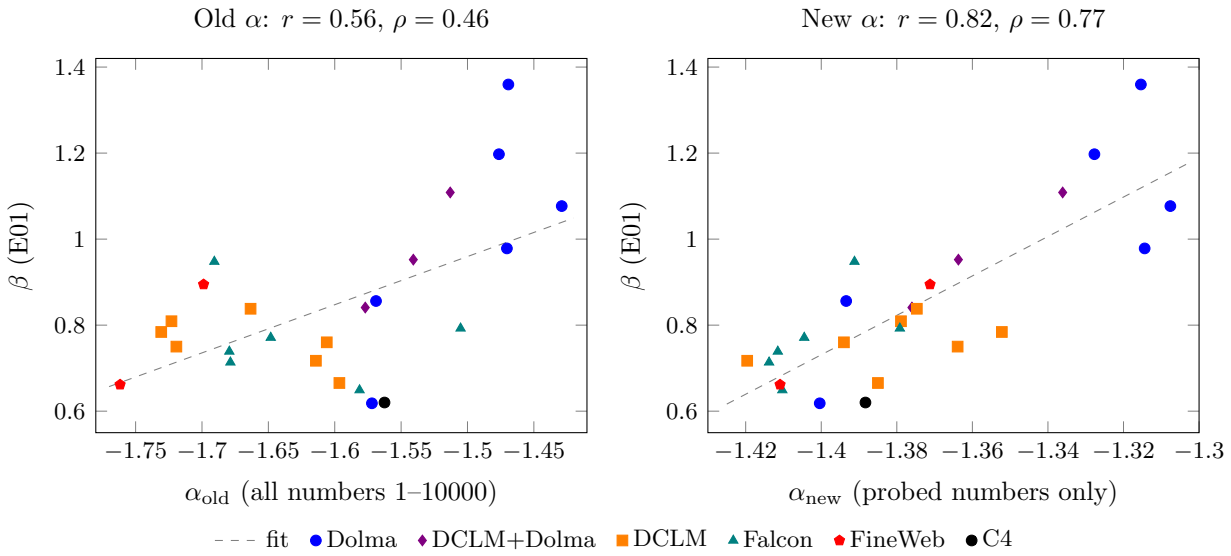


Figure 1: β against the two definitions of α; one point per model, colour = family. With the old α the trend comes mostly from Dolma (blue) versus the rest; with the new α the web-only families also follow the trend.

- It does not depend on any single model (leave-one-out: $r = 0.79$–0.84) or family.
- It survives removing Dolma ($r = 0.68$), holds within families ($r = 0.75$, vs 0.28 for α_{old}), and is strongest on the 8 independent base corpora ($r = 0.92$).
- The recipes are related (variants of the same corpus), so we also resampled whole families (family-level bootstrap): 95% interval for r is $[0.39, 0.90]$, which excludes 0. Using one average point per family: $r = 0.84$ ($p = 0.035$, 6 points).
- Controlling for code+math share weakens it: partial $r = 0.30$ ($p \approx 0.15$) for α_{new}, -0.23 for α_{old}. Code/math content explains part of the relation.

Caveats.

- **Chosen after seeing the data.** This α was proposed by a blind auditor (E10), but adopted

after the original α was known to correlate weakly. A second natural variant (pooling every probed number) agrees ($r = 0.79$, $\rho = 0.80$). It should be reported as exploratory and next to α_{old}, until tested on new models.

- Only 4 points per model, so α_{new} is a coarse slope.
- Without any code/math recipes the relation is positive but moderate ($\rho = 0.52$, $r = 0.44$, $p = 0.08$): the content confound is reduced, not removed.
- With E06's own layer choices the relation is weaker ($r = 0.42$), for the near-tie reason in E06; at E01's layers E06 gives $r \approx 0.70$.

What it addresses. Sensitivity to individual corpora, leave-one-out, the influence of Dolma, analysis by family, and uncertainty (VRQv Q2, H7EL, a1nj Q1). It also explains *why* the original α is weak: it measures mostly numbers β does not probe.

Appendix: all 25 models

Table 10: Per-model values, sorted by β. Layer and Order from E01; β (E01) $\pm$ s.d. over 3 prompt seeds; β (E06) = 10-group smooth fit at the same layer. *Recipe contains code or math documents.

Model	Family	Layer	Order	β (E01)	β (E06)	α_{old}	α_{new}
dolma1_7-no-reddit*	Dolma	8	0.91	1.36 ± 0.06	1.18	-1.47	-1.315
dolma1_7-no-flan*	Dolma	9	0.93	1.20 ± 0.06	1.19	-1.48	-1.328
dclm-baseline-25p-dolma1.7-75p*	DCLM+Dolma	8	0.93	1.11 ± 0.06	1.10	-1.51	-1.336
dolma1_6plus*	Dolma	9	0.92	1.08 ± 0.06	1.01	-1.43	-1.308
dolma1_7*	Dolma	6	0.93	0.98 ± 0.08	0.92	-1.47	-1.314
dclm-baseline-50p-dolma1.7-50p*	DCLM+Dolma	8	0.95	0.95 ± 0.05	0.83	-1.54	-1.364
falcon-and-cc-qc-tulu-10p	Falcon	9	0.93	0.95 ± 0.04	0.81	-1.69	-1.391
fineweb-edu	FineWeb	9	0.96	0.89 ± 0.04	0.87	-1.70	-1.371
dolma1_7-no-code*	Dolma	8	0.94	0.86 ± 0.06	0.77	-1.57	-1.393
dclm-baseline-75p-dolma1.7-25p*	DCLM+Dolma	10	0.94	0.84 ± 0.07	0.76	-1.58	-1.376
dclm-baseline-qc-7p-fw2	DCLM	8	0.95	0.84 ± 0.04	0.67	-1.66	-1.375
dclm-baseline-qc-fw-10p	DCLM	3	0.94	0.81 ± 0.03	0.76	-1.72	-1.379
falcon	Falcon	9	0.92	0.79 ± 0.05	0.68	-1.51	-1.379
dclm-baseline-qc-fw-3p	DCLM	9	0.93	0.78 ± 0.00	0.57	-1.73	-1.352
falcon-and-cc-qc-orig-10p	Falcon	8	0.94	0.77 ± 0.06	0.66	-1.65	-1.405
dclm-baseline-qc-20p	DCLM	9	0.95	0.76 ± 0.06	0.72	-1.61	-1.394
dclm-baseline-qc-7p-fw3	DCLM	9	0.92	0.75 ± 0.08	0.60	-1.72	-1.364
falcon-and-cc-qc-20p	Falcon	8	0.93	0.74 ± 0.06	0.68	-1.68	-1.411
dclm-baseline-qc-10p	DCLM	8	0.93	0.72 ± 0.06	0.66	-1.61	-1.420
falcon-and-cc-qc-10p	Falcon	8	0.92	0.71 ± 0.07	0.60	-1.68	-1.414
dclm-baseline	DCLM	8	0.90	0.67 ± 0.07	0.58	-1.60	-1.385
fineweb-pro	FineWeb	7	0.93	0.66 ± 0.01	0.58	-1.76	-1.411
falcon-and-cc	Falcon	9	0.93	0.65 ± 0.04	0.63	-1.58	-1.410
c4	C4	7	0.93	0.62 ± 0.05	0.57	-1.56	-1.388
dolma1_7-no-math-code	Dolma	4	0.94	0.62 ± 0.04	0.55	-1.57	-1.400